Beyazit Yalcinkaya

beyazit@berkeley.edu
people.eecs.berkeley.edu/~beyazit/
Cory Hall, Berkeley, CA 94720

Research Interests	Reinforcement learning, representation learning, multi-agent systems, simulation, statistical verification, runtime monitoring and assurance.
Software Packages	I am the primary developer and maintainer of the following Python packages: dfax, dfa-gym, and rad-embeddings (click on the package for more).
Highlighted Skills	Python, JAX, PyTorch, NumPy, C/C++, Haskell, and Prolog.

Education

2021 – 2027 (Expected)	**Ph.D. in Computer Science** Advised by Sanjit A. Seshia *University of California, Berkeley, CA, USA*
2015 – 2020	**B.Sc. in Computer Engineering** Valedictorian of the department (1st/249) *Middle East Technical University, Ankara, Turkiye*

Research Experience

2021 – Present	**Graduate Student Researcher** Multi-task reinforcement learning, supervised by Sanjit A. Seshia *University of California, Berkeley, CA, USA*
Summer 2026	**Research Intern** Multi-agent simulation, supervised by Marcell Vazquez-Chanlatte *Zoox, Foster City, CA, USA*
Summer 2023	**Applied Scientist Intern** Availability in distributed systems, supervised by Ankush Desai *Amazon Web Services, East Palo Alto, CA, USA*
Summer 2022	**Applied Scientist Intern** Observability in distributed systems, supervised by Ankush Desai *Amazon Web Services, Cupertino, CA, USA*
2017 – 2020	**Undergraduate Research Assistant** Timed automata and temporal logics, supervised by Ebru Aydin Gol *Middle East Technical University, Ankara, Turkiye*
Summer 2019	**Research Intern** Performance analysis of network functions, supervised by George Candea *École Polytechnique Fédérale de Lausanne, Lausanne, Switzerland*
Summer 2018	**Research Intern** Scheduling of real-time systems, supervised by Björn B. Brandenburg *Max Planck Institute for Software Systems, Kaiserslautern, Germany*

Teaching Experience

Spring 2026	**Graduate Student Instructor** CS 189/289A Introduction to Machine Learning *University of California, Berkeley, CA, USA*
Fall 2022	**Graduate Student Instructor** EE/CS 149/C249A Introduction to Embedded Systems *University of California, Berkeley, CA, USA*
Fall 2017	**Undergraduate Teaching Assistant** CENG 230 Introduction to C Programming *Middle East Technical University, Ankara, Turkiye*

Publications

[1] Abhinav Pomalapally, Arya Raeesi, Kevin Kai-Chun Chang, **Beyazit Yalcinkaya**, and Sanjit A. Seshia, *Scenario-Based Compositional Statistical Model Checking for Safety Specifications*, **International Conference on Runtime Verification (RV)**, 2026.

[2] **Beyazit Yalcinkaya**, Marcell Vazquez-Chanlatte, Ameesh Shah, Hanna Krasowski, and Sanjit A. Seshia, *Automata-Conditioned Cooperative Multi-Agent Reinforcement Learning*, **International Conference on Machine Learning (ICML)**, 2026.

[3] **Beyazit Yalcinkaya**, Marcell Vazquez-Chanlatte, and Sanjit A. Seshia, *Learning Task-Conditioned Policies with Natural Language and Graph Embeddings of Formal Specifications*, **International Conference on Neuro-symbolic Systems (NeuS)**, 2026.

[4] Pau de las Heras Molins, **Beyazit Yalcinkaya**, Lasse Peters, David Fridovich-Keil, and Georgios Bakirtzis, *Controllability in Preference-Conditioned Multi-objective Reinforcement Learning*, **International Conference on Neuro-symbolic Systems (NeuS)**, 2026.

[5] **Beyazit Yalcinkaya**, Niklas Lauffer, Marcell Vazquez-Chanlatte, and Sanjit A. Seshia, *Provably Correct Automata Embeddings for Optimal Automata-Conditioned Reinforcement Learning*, **International Conference on Neuro-symbolic Systems (NeuS)**, 2025.

[6] **Beyazit Yalcinkaya***, Niklas Lauffer*, Marcell Vazquez-Chanlatte*, and Sanjit A. Seshia, *Compositional Automata Embeddings for Goal-Conditioned Reinforcement Learning*, **Conference on Neural Information Processing Systems (NeurIPS)**, 2024. *Equal contribution.

[7] **Beyazit Yalcinkaya***, Niklas Lauffer*, Marcell Vazquez-Chanlatte, and Sanjit A. Seshia, *Automata-Conditioned Reinforcement Learning with Experience Replay*, **NeurIPS 2023 Workshop on Goal-Conditioned Reinforcement Learning**, 2023. *Equal contribution.

[8] **Beyazit Yalcinkaya**, Hazem Torfah, Daniel J. Fremont, and Sanjit A. Seshia, *Compositional Simulation-Based Analysis of AI-Based Autonomous Systems for Markovian Specifications*, **International Conference on Runtime Verification (RV)**, 2023.

[9] **Beyazit Yalcinkaya**, Hazem Torfah, Ankush Desai, and Sanjit A. Seshia, *ULGEN: A Runtime Assurance Framework for Programming Safe Cyber-Physical Systems*, **IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)**, 2023.

[10] Niklas Lauffer*, **Beyazit Yalcinkaya***, Marcell Vazquez-Chanlatte, Ameesh Shah, and Sanjit A. Seshia, *Learning Deterministic Finite Automata Decompositions from Examples and Demonstrations*, **Formal Methods in Computer-Aided Design (FMCAD)**, 2022. *Equal contribution.

[11] Mert Ergurtuna, **Beyazit Yalcinkaya**, and Ebru Aydin Gol, *An Automated System Repair Framework with Signal Temporal Logic*, **Acta Informatica**, 2022.

[12] **Beyazit Yalcinkaya** and Ebru Aydin Gol, *Clock Reduction in Timed Automata while Preserving Design Parameters*, **International Conference on Formal Methods in Software Engineering (FormaliSE)**, 2019.

[13] **Beyazit Yalcinkaya**, Mitra Nasri, and Björn B. Brandenburg, *An Exact Schedulability Test for Non-Preemptive Self-Suspending Real-Time Tasks*, **Design, Automation and Test in Europe Conference (DATE)**, 2019.